

A Density Regression Refiner Framework with Pyramidal Feature Fusion Networks for Crowd Counting: Supplementary Material

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Abstract—Density regression-based crowd counting estimates crowd size (the number of individuals) by predicting a density map whose integral gives the total crowd count. While recent CNN and hybrid models achieve high accuracy, many are computationally expensive and unsuitable for real-time edge deployment. We propose a density regression refiner framework that balances accuracy and efficiency through a modular design. It integrates the following three components: (a) a lightweight backbone selected from the MobileNet or ShuffleNet series for multi-scale feature extraction, (b) a CBAM-based feature refiner module (FRM) for channel-spatial refinement/enhancement, and (c) a pyramidal feature fusion module (PFFM), implemented using FPN, PANet, or BiFPN, for structured multi-scale feature aggregation. A point-wise convolutional layer serves as a lightweight density regressor to generate the final density map. Extensive experiments have been conducted on crowd counting benchmarks: ShanghaiTech, UCF_CC_50, and UCF-QNRF. Experiments demonstrate the competitive performance (MAE/MSE) of this proposal with fewer model parameters and lesser number of floating-point operations (FLOPs).

Index Terms—Crowd Counting, Density Estimation, Lightweight CNN, Attention Mechanism, Multi-Scale Feature Fusion.

I. GROUND TRUTH GENERATION

This work adopts the approach proposed in [1] and utilizes the geometry-adaptive kernels to handle the extremely crowded scenes. To create the ground truth, we apply a Gaussian kernel (normalized at 1) for blurring each head annotation, taking into account the spatial dispersion across all images in each dataset. The parameters associated with the ground truth generation are presented in Table I.

Assuming an annotated head is present at pixel x_i , this can be represented as a Dirac delta function:

$$\delta(x - x_i) = 1 \quad (1)$$

An image containing N annotated heads can then be expressed by the following function:

$$H(x) = \sum_{i=1}^N \delta(x - x_i) \quad (2)$$

TABLE I

THE GROUND TRUTH GENERATION METHODS FOR VARIOUS DATASETS.

Dataset	Generating Method
ShanghaiTech Part_A [1]	Geometry-adaptive kernels
UCF_CC_50 [2]	
ShanghaiTech Part_B [1]	Fixed Gaussian kernel: $\sigma = 15$
UCF-QNRF [3]	Fixed Gaussian kernel: $\sigma = 4$

The geometry-adaptive kernel is described as:

$$F(x) = \sum_{i=1}^N \delta(x - x_i) * G_{\sigma_i}(x), \quad \text{where } \sigma_i = \beta \bar{d}_i \quad (3)$$

In the ground truth δ , we utilize $\bar{d}_i$ to represent the average distance between a head annotation and the k nearest head annotations for each target object x_i . We apply the convolution of $\delta(x - x_i)$ with a Gaussian kernel using parameter σ_i (standard deviation), where x denotes the pixel location in an image, for generating the density map. In our experiments, we adopt the configuration from [1] with $\beta = 0.3$ and $k = 3$. We adjust the Gaussian kernel to the typical head size to blur all annotations uniformly in the sparse crowd scenarios. The specific configurations for various datasets are presented in Table I.

II. EVALUATION METRICS

The Mean Absolute Error (MAE) and Mean Squared Error (MSE) are utilized for evaluation, defined as follows:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |c_i - \hat{c}_i| \quad (4)$$

$$\text{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N |c_i - \hat{c}_i|^2} \quad (5)$$

where N represents the number of testing images, c_i denotes the ground truth count, and $\hat{c}_i$ denotes the estimated (predicted) count which is described in the following manner:

$$\hat{c}_i = \sum_{h=1}^H \sum_{w=1}^W p_{h,w} \quad (6)$$

where H and W represent the height and width of the density map, respectively, while $p_{h,w}$ denotes the pixel at position (h, w) of the estimated (generated) density map, while $\hat{c}_i$ signifies the estimated count for image X_i . The total count of individuals is obtained by integrating the estimated (predicted) density map or summing the pixel values of the density image.

MAE calculates the mean absolute difference between the ground truth (actual) and estimated (predicted) counts, while MSE computes the mean of the squared differences. MAE denotes the counting accuracy and MSE denotes the robustness of our proposed model.

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